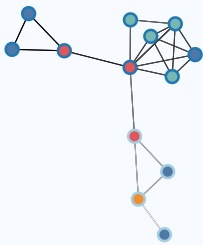


SERGIO RUIZ-CARMONA

I work as a [Clinical Data Analyst](#), uncovering insights to drive informed decision-making. Proficient in extracting EMR records using SQL, I specialize in analyzing and visualizing data using R and Power BI. With a deep understanding of clinical data and a research background, I bring a unique perspective to optimizing healthcare outcomes.

🏠 **Current address:** 1/144 Brunswick Rd, Brunswick (Victoria 3056), Australia
📦 **Relocating to Barcelona in March 2025**



View this resume [online](#)

SELECTED EDUCATION

Jan 2017
|
Jul 2012

●

PhD in Biomedicine
Universitat de Barcelona

Barcelona, ES

- Virtual Screening for novel MoA: Apps. and method developments
- Development and application of **structure-based drug discovery** methods
- Awards: **Extraordinary prize** for PhD Thesis and **Ramon Margalef award** for best publication

WORK HISTORY

Present
|
Oct 2022

●

Clinical Data Analyst
Centre for Health Analytics
Royal Children’s Hospital

Melbourne, AUS

- I work as a Clinical Data Analyst, uncovering insights to drive informed decision-making. Proficient in extracting Epic EMR records using SQL, I specialize in analyzing and visualizing data using R and Power BI.

Oct 2022
|
Jan 2019

●

Senior Research Officer and Bioinformatician
Inouye Lab
Baker Heart and Diabetes Institute

Melbourne, AUS

- I worked in the intersection between Structural Biology and Genomics, trying to understand how rare missense variants alter protein structures and can induce disease

Dec 2019
|
Jan 2017

●

Postdoctoral Researcher
Barril Lab, Universitat de Barcelona

Barcelona, ES

- After finishing my PhD, I worked in exciting collaborative projects in the field of epigenetics and cancer, where I carried out multidisciplinary research and learned new experimental skills

SKILLS AND TRAINING

●

Computational Tools
Data visualization, Data analysis, Dashboards (Power BI, R Shiny), CI/CD, Machine learning, Statistical analysis, REDCap, Bioinformatics and Drug Discovery

●

Programming
R, Python, SQL, Perl, C++, LaTeX, Bash, Java, HTML/CSS and Android and Web Development

●

Management Skills
2021 EMBO Practical Course: Research to service: Planning and running a bioinformatics core facility; Mental Health and OHS certifications

CONTACT

✉ sruizcarmona@gmail.com
🌐 [sruizcarmona](#)
in [Sergio Ruiz](#)
🌐 [ruizsergio.com](#)
📞 +610478759880

PROGRAMMING



HIGHLIGHTS

Highly experienced in:

- Computational biology and drug design
- Bioinformatics and biostatistics
- Machine Learning
- Data visualization
- Experimental techniques

Motivated and **collaborative**, recently involved in **Machine Learning**, NGS data analysis, **UK Biobank** and genomics prediction

MORE INFO

See full CV at cv.ruizsergio.com for a complete list of positions, publications and more

Reference contact details provided upon request



Source code available in [github](#)
Last updated in Jan, 2025

- **Certifications**
Certified Health Informatician Australasia (CHIA), UX/UI Design (RMIT), Dataviz with R (IBM), Machine Learning (IBM)

OUTREACH AND OTHER ACTIVITIES

- Sep 2021 | Sep 2020 ● **Australasian Leadership Computing Grants**
Story about NCI Computing Grant
 - Together with Mike Inouye, we were awarded a 1 year computing grant by the Australian National Computational Infrastructure to study COVID-19 proteins and possible drug treatments
- Sep 2021 ● **New tools for new medicines, The Conversation [article](#)**
Lay-summary of some of my work and opinion for The Conversation Spain
- Apr 2021 ● **Shiny App development**
Related to a project we published in the Journal *Circulation*
- Dec 2020 ● **COVID-19 daily [dashboard](#), until Nov 2020**
Personal project to showcase COVID-19 evolution in Victoria during major lockdown in 2020

MAIN SCIENTIFIC OUTPUT

- Nov 2023 ● **COVID-19 thromboembolism is reduced in ambulatory, but not hospitalized patients, following COVID-19 vaccination**
Hematologica
Read it [here](#)
 - Hannah Stevens *, **Sergio Ruiz-Carmona** *, Karlheinz Peter and James D McFadyen
- Aug 2023 ● **Semiautomatic Training Load Determination in Endurance Athletes**
Journal for the Measurement of Physical Behaviour
Read it [here](#)
 - Christophe Dausin *, **Sergio Ruiz-Carmona** *, [...] and Guido Claessen
- Apr 2021 ● **Oxygen Pathway Limitations in Patients with Chronic Thromboembolic Pulmonary Hypertension**
Circulation
Read it [here](#)
 - Erin J Howden *, **Sergio Ruiz-Carmona** *, [...] Andre La Gerche, Marion Delcroix and Guido Claessen
 - Result of a Bioinformatics Core collaboration. [Shiny app](#) developed
- Jan 2017 ● **Dynamic undocking and the quasi-bound state as tools for drug discovery**
Nature Chemistry
Read it [here](#)
 - **Sergio Ruiz-Carmona**, Peter Schmidtke, [...] Rod Hubbard and Xavier Barril
 - Highlighted in its [issue cover](#)
- Sep 2014 ● **rDock: a fast, versatile and open source program for docking ligands to proteins and nucleic acids**
PLoS Computational Biology
Read it [here](#)
 - **Sergio Ruiz-Carmona**, Daniel Alvarez-Garcia, [...] Xavier Barril, Rod Hubbard and S David Morley

EVERYDAY TOOLBOX

- RStudio
- Data viz & Shiny dashboards
- Bash, ssh & GitHub (local or HPC)
- ML optimization (genetic algorithms, MC, ...)
- Linear models & random forests

WEB

I have also developed [rDock](#) and [Barril Lab](#) websites, used as dissemination platforms where the different tutorials and blog entries are shared with the community



I have been involved in more than 20 collaborative projects in very different fields. I enjoy helping other researchers and collaborators to work in common scientific challenges



My ultimate goal is to help people: from drug discovery to precision medicine and hospital care data analysis



Part of my research in drug discovery became a [drug patent](#) for cancer treatment, currently under development at [Oniria Therapeutics](#)